

Natural Language Processing Transformers

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CSC525 Principles of Machine Learning

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May 18, 2025

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Thinking back to 2019, the natural language processing (NLP) field was at a turning point. It was the first year we truly saw the transformative potential of a new kind of model called a transformer. Models like BERT, short for Bidirectional Representations from Transformers had just begun reshaping how we approached language understanding, while tools like GROVER provided early insights into the dual nature of NLPs potential of generating incredibly convincing fake news, and then detecting that an article was AI-generated. At the time, transformer-based models felt groundbreaking. Before 2019, NLP largely relied on simpler architectures like recurrent neural networks (RNNs) and convolutional neural networks (CNNs), which handled language step by step or, through fixed or through fixed context window sizes. Transformers introduced a powerful concept known as self-attention, enabling the model to understand context within sentences much more effectively (Vaswani et al., 2017.) BERT took advantage of this idea to pre-train on much larger amounts of text data to absorb the nuances of natural language, and then fine-tune quickly for various tasks. Before, the approach was revolutionary enough that tech writers like Latysheva (2019) declared it “the year of BERT”, emphasizing how transfer learning would soon dominate the field of NLP research.

Looking back from 2025, it’s striking how prescient those early discussions were. The initial enthusiasm around BERT and transformer architectures wasn’t just hype. BERT laid the groundwork for extraordinary progress. In healthcare, NLP has evolved far beyond its early clinical documentation applications. Today, advanced transformer-based systems routinely analyze patient conversations in real-time to detect subtle indicators of mental health conditions, enabling earlier interventions than previously possible (Martínez-Castaño et al., 2021). Likewise, education has witnessed transformative tools, like intelligent tutoring systems tailoring

expectations to a student's unique conversational patterns (Arnau-González, Solera-Monforte, Wu, & Arevalillo-Herráez, 2025), something that was merely theoretical in 2019.

In the broader AI landscape, transformers have blurred boundaries between disciplines. Reflecting on 2019, we remember transformer models in primarily text-based tasks, but by 2025, they've fully integrated into multimodal models like ChatGPT, Gemini, and Claude (Adhikari, 2025.) Such models have exemplified how transformer architectures enable machine learning across domains, transferring knowledge from one data type or modality to another seamlessly. The interconnectedness of AI subfields was aspirational in 2019, today, it's a practical reality.

Alongside technological leaps, we've seen ethical challenges escalate as well. In 2019, the GROVER paper (Zellers et al, 2020) raised concerns over deepfake texts, convincing machine-generated articles that are indistinguishable from human writing. GROVER itself served as both an alarming demonstration and a potential solution, capable of detecting its own fake outputs. Yet, in the intervening years, we've faced unprecedented issues with AI-generated disinformation campaigns, it seems a new threat is identified weekly in our modern times (Vicens, 2025.) The capabilities demonstrated by GROVER were only the beginning. Today's models are not only more convincing, but also far easier for individuals to misuse. This evolution has forced governments, researchers, and tech companies to actively collaborate on solutions such as watermarking AI-generated content and deploying transformer-based detectors at scale. From a technical standpoint, the ML methods we once considered groundbreaking have evolved substantially. The practice of fine-tuning pretrained transformers, once a dream, has become a foundation. By 2025, we've pushed this concept much further. Transformers now dynamically adapt their knowledge to entirely new domains without extensive retraining. Additionally, attention mechanisms have become increasingly sophisticated. Techniques like

retrieval-augmented generation (RAG), first explored shortly after 2019, now have models to draw seamlessly on external databases and factual sources, dramatically enhancing accuracy and reducing hallucinations.

Perhaps most strikingly, the massive growth in model size seen after BERT has given way to a renewed focus on efficiency and sustainability. In 2019, the idea of scaling models to billions of parameters was awe-inspiring. Today, efficiency-oriented architectures like DistilBERT and TinyBERT have proven that smaller, cleverly optimized models can achieve nearly identical performance, making powerful NLP performance accessible on everyday devices without heavy computational burdens. To get a feel for this, I used Hugging Face to run DistilBERT and perform sentiment analysis for Amazon listings. The experiment underscored how effectively these optimized models can provide quick, accurate insights even on modest hardware like my NVIDIA 3080 GPU. This would have seemed remarkable back when transformer models were first emerging, today, it feels slightly underwhelming.

Looking back to 2019, it's clear we stood at the dawn of a profound shift. The enthusiasm around BERT and GROVER wasn't just excitement about technological novelty. It signaled the arrival of technologies capable of reshaping society. Looking forward from 2025, the possible future applications of NLP seem limitless. Beyond today's advances in healthcare and education, NLP technology is beginning to redefine personal assistants, using agentic AGI to make them not merely responsive, but genuinely insightful companions capable of understanding and anticipating complex human needs. We might soon see NLP playing a pivotal role in mental health therapy, with emotionally intelligent conversational agents providing preliminary screenings or ongoing emotional support, accessible even in underserved regions. Another exciting area is augmented reality, where NLP can help create truly immersive interactions.

Imagine simply describing a scene, and your AI-driven AR glasses instantly render it visually overtop your real-world AR systems.

The transformative potential of NLP doesn't stop at human language. It's reshaping other AI disciplines as well. Transformers designed initially for text have already shown tremendous success in fields like computer vision and robotics. Techniques like self-attention, first popularized by NLP, have inspired breakthroughs in visual recognition, enabling more nuanced understanding of images and video. Moreover, NLP's transfer-learning techniques have influenced how AI researchers approach everything from autonomous vehicles to protein folding prediction (Vieira, Handojo, & Wilke, 2024). Models trained first on general-purpose data sets now routinely specialize across entirely unrelated tasks, reducing the resources needed to build sophisticated transfer learning AI systems.

As exciting as these possibilities are, the ethical and social implications remain troubling. GROVER, introduced in 2019, offered us an early warning about AI-generated misinformation. Today, concerns about convincing AI-generated propaganda aren't merely hypothetical, they're a daily reality, challenging democratic institutions worldwide (Rêgo & Weber, 2025.) Misinformation isn't our only concern. Advanced NLP agents can impersonate individuals convincingly, making identity verification more challenging and potentially undermining trust online. Bias encoded in training data continues to be a pressing issue. Unless carefully managed, NLP systems inadvertently reinforce harmful stereotypes, marginalize vulnerable communities, or violate privacy through data leaks. This underscores an urgent need for clearer regulations, transparent models, and ethical AI development standards.

From a technical perspective, NLP has been built on a powerful foundation of ML innovations. Initially, it was transfer learning and self-attention mechanisms that revolutionized NLP in 2019. Today, methods like RAG hold enormous promise. By dynamically querying external knowledge sources during text generation, these methods have substantially reduced factual inaccuracies like hallucinations, where the model confidently outputs incorrect or misleading information. We're also seeing promising results from reinforcement learning with human feedback, helping align NLP systems more closely with human values and expectations. Looking forward, even more intriguing possibilities lie in areas like neuro-symbolic AI, which combines traditional symbolic reasoning with neural networks to produce more interpretable and controllable language models.

Ultimately, reflecting back from our view in 2025, the story of NLP since 2019 has been one of extraordinary advancement tempered by sobering ethical challenges. It's clear we are still navigating a period of remarkable potential and significant responsibility. As NLP continues evolving, the critical task facing researchers will be ensuring these powerful technologies serve humanity rather than control or mislead it. Our experience over these past few transformative years provides hope that we are increasingly equipped to meet that challenge thoughtfully and responsibly.

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